Lab Sheet 05 – Question 02: VTV Cable TV System

SE1020 – Object Oriented Programming

(a) Nouns that can be potential classes

VTV, company, TV channels, channel, person, customer, registration, customer details, package, customer number, customer table, subscription table, channel details table, payments table, subscription, bill, website, online tasks, list of channels, weekly TV schedule, TV schedule, email, credit card, credit card details, amount, payment, monthly bills, data, system, month, week.

(b) Rejecting nouns using the five rules

Noun	Decision	Rule applied / Reason
VTV, company	Rejected	Irrelevant – the company is the system owner, not an object the system models
person	Rejected	Redundant – same as Customer
registration	Rejected	Irrelevant/Event – it is an operation (register) done by Customer
customer details	Rejected	Attribute – name, address, email etc. belong to Customer
customer number	Rejected	Attribute of Customer
package	Rejected	Redundant – same concept as Subscription (selected channels)
customer table, subscription table, channel details table, payments table	Rejected	Implementation constructs – database tables, not real-world classes
website, system, online tasks	Rejected	Implementation constructs / Irrelevant
data, month, week	Rejected	Irrelevant / Vague
list of channels	Rejected	Attribute – collection held by Subscription

Noun	Decision	Rule applied / Reason
weekly TV schedule	Rejected	Adjective class – "weekly" only describes TV schedule (same as TV Schedule)
monthly bills	Rejected	Adjective class – "monthly" only describes Bill
email	Rejected	Implementation construct / attribute of Customer
credit card details	Rejected	Redundant – same as Credit Card
amount	Rejected	Attribute of Payment
TV channels	Rejected	Redundant – same as Channel

Final set of classes

1. **Customer**
2. **Subscription**
3. **Channel**
4. **TVSchedule**
5. **Bill**
6. **Payment**
7. **CreditCard**

(c) CRC Cards (5 classes)

Card 1 – Customer

Responsibilities	Collaborators
Register with VTV and store customer details	Subscription
Provide unique customer number	Bill
Login to the website	Payment
Add channels to subscription	Channel
Remove channels from subscription	
Request weekly TV schedule by email	

Responsibilities	Collaborators
Pay monthly bill	

Card 2 – Subscription

Responsibilities	Collaborators
Store package (channels selected by customer)	Customer
Add a channel	Channel
Remove a channel	
Store list of channels for weekly schedule emails	

Card 3 – Channel

Responsibilities	Collaborators
Store channel details (id, name, price)	Subscription
Provide channel price for billing	TVSchedule
Provide TV schedule of the channel	Bill

Card 4 – Bill

Responsibilities	Collaborators
Generate monthly bill	Customer
Calculate total using subscribed channels	Subscription
Send bill to customer	Channel
Keep paid / unpaid status	Payment

Card 5 – Payment

Responsibilities	Collaborators
Record payment amount and date	Customer
Accept credit card details	CreditCard
Store payment details	Bill

Responsibilities	Collaborators
Update bill status after payment	